

Cutaneous dysbiosis characterizes the post-allogeneic hematopoietic stem cell transplantation period

Adela R. Cardones,^{1,*} Akintunde Emiola,^{2,*} Russell P. Hall III,³ Anthony D. Sung,^{4,5} Jennifer Zhang,³ Amy J. Petty,³ Charles Puza,⁶ Lauren M. Bohannon,⁴ Amy T. Bush,⁴ Meagan V. Lew,⁴ Elizabeth Fleming,² Yingai J. Jin,³ Krista Rowe Nichols,⁴ Vaibhav Jain,⁶ Simon G. Gregory,^{7,8} Keith M. Sullivan,^{4,5} Nelson J. Chao,^{4,5} and Julia Oh^{3,9}

¹Division of Dermatology, Department of Internal Medicine, University of Kansas Medical Center, Kansas City, KS; ²Microbial Therapeutics Unit, National Institute of Dental and Craniofacial Research, National Institutes of Health, Bethesda, MD; ³Department of Dermatology, ⁴Division of Hematologic Malignancies and Cellular Therapy, Department of Medicine, and ⁵Duke Cancer Institute, Duke University Medical Center, Durham, NC; ⁶Duke University School of Medicine, Durham, NC; ⁷Duke Molecular Physiology Institute and ⁸Department of Neurology, Duke University School of Medicine, Durham, NC; ⁹Department of Integrative Immunobiology, Duke University School of Medicine, Durham, NC

Key Points

- The cutaneous microbiome in allogeneic HSCT patients is enriched for pathobionts.
- The nature and clinical implications of host–microbiome interaction in the skin of allogeneic HSCT patients need further characterization.

Gut dysbiosis is linked to mortality and the development of graft-versus-host disease after hematopoietic stem cell transplantation (HSCT), but the impact of cutaneous dysbiosis remains unexplored. We performed a pilot observational study, obtained retroauricular and forearm skin swabs from 12 adult patients before conditioning chemotherapy/radiation and at 1 week, 1 month, and 3 months after allogeneic HSCT, and performed shotgun metagenomic sequencing. The cutaneous microbiome among HSCT patients was enriched for gram-negative bacteria such as *Escherichia coli* and *Pseudomonas*, fungi, and viruses. Enrichment with bacteriophages and *Polyomavirus* species was observed among patients who died within 1 year. We observed longitudinal stability of the cutaneous microbiome at the 3-month time point among those who survived beyond 1 year after HSCT, although these may simply be a reflection of the overall medical status of the patients. There was no association with fungal abundance and any of the outcomes observed. The cutaneous microbiome may be a reservoir of pathobionts among allogeneic HSCT patients. Our findings suggest that cutaneous dysbiosis exists after HSCT, but the ultimate implication of this to patient outcomes remains to be seen through larger studies.

Introduction

Skin graft-versus-host disease (GVHD) affects up to ~70% to 80% of allogeneic hematopoietic stem cell transplantation (HSCT) patients,¹ and yet, the underlying pathogenesis is poorly understood. Abnormalities in the gut microbiome have been associated with the development of GVHD² and mortality,³ but the impact of the skin microbiome on HSCT outcomes is unknown. Chronically immunosuppressed patients have a disrupted skin microbiome,⁴ and skin dysbiosis has been linked to inflammatory cutaneous disorders,⁵ facilitating Th17 polarization,⁶ and skin disease progression.⁵ Skin commensals are critical for maintaining local immune homeostasis, keeping pathogenic organisms in check,⁷ and are necessary for the development of the skin's regulatory T-cell compartment.⁸ Based on this, we hypothesized that the skin microbiome is altered among allogeneic HSCT patients and might

Submitted 19 March 2021; accepted 17 December 2024; prepublished online on *Blood Advances* First Edition 24 January 2025; final version published online 28 April 2025. <https://doi.org/10.1182/bloodadvances.2021004792>.

*A.R.C. and A.E. contributed equally to this study.

Data sets in this manuscript are available in NCBI (National Center for Biotechnology Information) repositories (accession numbers GSE190338 and PRJNA1198494).

Data are available upon reasonable request from the corresponding author, Adela R. Cardones (acardones@kumc.edu).

The full-text version of this article contains a data supplement.

Licensed under [Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International \(CC BY-NC-ND 4.0\)](#), permitting only noncommercial, nonderivative use with attribution.

be an important biomarker for outcomes. We also hypothesized that skin dysbiosis drives the innate, inflammatory response characteristic of acute GVHD (aGVHD) of the skin, and keratinocytes are active mediators of this inflammation. Therefore, we aimed to describe the diversity and composition of the cutaneous microbiome among allogeneic HSCT patients across the 90-day post-transplant recovery period. We obtained data on the development of aGVHD as well as mortality.

Methods

After obtaining approval from the Duke Institutional Review Board and written informed consent, we obtained skin swabs from the forearm (arm, $n = 61$) and retroauricular areas (ear, $n = 63$) of 12 adult patients (supplemental Table 1) and from 4 primary caregivers who consented to participate in the study to identify any potential microbial transmission patterns or risk reservoirs between caregivers and patients. Skin swabs from HSCT patients were obtained before conditioning chemotherapy/radiation and at 1 week, 1 month, and at 3 months after allogeneic HSCT, corresponding to landmarks in the immune recovery process after HSCT. The sites were selected to represent oily and dry skin and body regions commonly affected by aGVHD. We had a maximum of 1 sample from each body site and time point, with some missing samples. Shotgun metagenomic sequencing of samples was performed as previously described.⁹ Patient characteristics and outcomes, development of acute skin GVHD, and 1-year mortality were collected (supplemental Table 1). Other pertinent clinical characteristics such as the development of systemic infections and antibiotic exposure were also collected (supplemental Table 2). We had previously used archival, formalin-fixed aGVHD skin biopsy tissue from an independent set of patients to confirm the expression of key antimicrobial cytokines S100A8 and IL36 by immunohistochemistry and immunofluorescence in keratinocytes of patients with aGVHD (data not shown). Because keratinocytes are more likely to be in contact with the cutaneous microbiome and are capable of mounting an inflammatory response, we performed single-cell RNA sequencing (scRNA-seq) on the epidermal compartment from a patient who developed skin aGVHD and compared it with that of a healthy volunteer. Anonymized truncal skin from a healthy adult was obtained through an institutional review board–approved tissue bank protocol and was used as healthy control for scRNA-seq and immunohistochemistry.

Skin swab processing and skin microbiome analysis

Reads processing and taxonomic characterization of the skin microbiome. We removed adapter sequences from raw metagenomics reads using Trim Galore¹⁰ and discarded human reads by mapping to hg19 genome using Bowtie2.¹¹ Low-quality reads were discarded using Sickle.¹² For taxonomic annotation, we mapped cleaned metagenomic reads to PanDB¹³ and reassigned ambiguous reads using Pathoscope 2.¹⁴ We compared the microbial composition of the skin microbiome in our cohort with a previously published data set from healthy volunteers.⁹ We compared the microbiome characteristics between patients in our cohort who developed acute cutaneous GVHD and those who did not, as well as between patients who survived 1 year after HSCT and those who did not.

ARG analyses. To identify plasmid-borne antimicrobial resistant genes (ARGs), we began by concatenating all samples and assembled sequence reads into contigs using MEGAHIT 1.0.5.¹⁵ We discarded contigs <1 kb and predicted plasmids' contigs using PlasFlow.¹⁶ Next, we used Prodigal¹⁷ to predict genes in plasmids and subsequently identified ARGs using DeepARG.¹⁸ We mapped each sample to the pool of ARGs using Bowtie2 and reassigned ambiguous reads using Pathoscope2.¹⁴

Growth rate analysis of *Cutibacterium acnes* strains. We estimated subspecies/strain-level growth rate using strain-level metagenomic estimation of growth rate.¹⁹ We retrieved 187 strains of *C. acnes* from NCBI (National Center for Biotechnology Information) GenBank and constructed the species database using a single-nucleotide polymorphism (SNP) assignment threshold of 0.6, without iterative clustering, resulting in 8 clusters (supplemental Table 2). Because strains of *C. acnes* have low within-species genetic diversity, we detected clusters at a 0.5× coverage cutoff and reported growth rates for clusters with coverage ≥1× and nonzero SNP count ≥100 in a given sample, as described previously.¹⁹ We used a cluster detection threshold of 0.2 to identify subspecies clusters in samples.

Virus analyses. We downloaded 760 453 viruses or viral fragments and their annotations from integrated microbial genomes/virus²⁰ to create our viral database. We mapped each sample to the database using Bowtie2,¹¹ reassigned ambiguous reads using Pathoscope2,¹⁴ and excluded viruses present in <3 samples. This resulted in hits to 17 541 viruses/viral fragments annotated as bacterial or eukaryotic viruses. We performed additional analysis for polyomavirus, because we had a previous analysis (not shown) that suggested enrichment for polyomavirus but not for others, such as human papilloma viruses, in the skin of HSCT patients. Polyomaviruses are relatively limited in types compared with other cutaneous viruses and yet have relatively high prevalence and can persist in the skin of healthy individuals.²¹ In addition, cutaneous polyomaviruses can cause significant cutaneous disease in immunosuppressed individuals. For strain-level polyomavirus analyses, we downloaded all sequences annotated as “human polyomavirus” ($n = 20$) from NCBI GenBank and generated a phylogenetic tree using FastTree,²² based on whole-genome alignment generated using mafft.²³

To investigate polyomavirus strain transmission in a pair of samples, we selected the most abundant strain (HM011556.1), with coverage >10× in most samples. First, we mapped each sample to HM011556.1 and retrieved polymorphic nucleotide coverage at each nucleotide position using Samtools (mpileup)²⁴ and a custom script. For every combination pair, we calculated the expected probability (P) for a nucleotide (n) at a given position with

$$P_n = \frac{n_{\text{sample1}} + n_{\text{sample2}}}{t_{\text{sample1}} + t_{\text{sample2}}},$$

in which n represents nucleotide coverage and t , total coverage. Next, we performed multinomial test (using Monte Carlo approach) at each position for each sample, using the nucleotide coverage and probabilities as input. A nucleotide position was considered an SNP if either sample pair had P values <.05 at that position.

Statistics. We conducted all statistical analyses using R software. To calculate P values between groups for repeated measures, we used unrestricted permutation implemented using a custom R script.

We first computed a test statistic from an observation and generated 500 permutations by shuffling the dependent variable. The F statistic was used as the test statistic in the permutation for analysis of variance. We expressed the P value as the proportion of permutations resulting in a larger test statistic than the observed statistics. P values were adjusted under the Benjamini-Hochberg procedure.

Single-cell isolation, RNA-seq library preparation, and sequencing of skin biopsy samples

After 2 brief washes in cold phosphate-buffered saline, the freshly obtained skin biopsies were incubated in 4-mL dispase (1 unit/mL; Thermo-Fisher Scientific, Waltham, MA) at 4°C overnight. Next day, the epidermis and dermis were separated from each other and chopped into small pieces with a pair of scissors. The epidermal pieces were digested with 0.25% trypsin for 15 minutes, and the dermal pieces were digested with collagenase B (1 mg/mL; Sigma, St. Louis, MO) for 1 hour, followed by the addition of 0.25% trypsin and incubation for another 15 to 30 minutes. During digestion, the tubes are placed at 37°C water bath with agitation at 240 revolutions per minute. The digestion mixtures were neutralized in 10% fetal bovine serum in Dulbecco's modified Eagle medium and filtered through 70- μ m and then 40- μ m strainers. The cell suspension was centrifuged at 300g for 5 minutes to obtain cell pellets, washed 3 times, and then resuspended in 0.04% bovine serum albumin at 1000 cells per μ L. After quality check, cells were immediately used for complementary DNA amplification and chromium library construction using scRNA-seq library kit v2, according to the manufacturers' recommendations (10X Genomics) by the Molecular Genomics Core facility at the Duke Molecular Physiology Institute. Complementary DNA libraries were sequenced on the Illumina HiSeq 4000, with a read length of 150 bp by the Duke Center for Genomics and Computational Biology core facility.

Analysis of scRNA-seq data. Analysis of the scRNA-seq data followed the manufacturers' protocol (10X Genomics). Briefly, demultiplexed raw base call files were generated by Illumina sequencers into FASTQ files, and alignment was performed to the h19 human transcriptome. Filtering, barcode, and unique molecular identifier counting were performed using Cell Ranger software version 3.1 and default parameters (10X Genomics). Secondary statistical analysis was performed using the R package Seurat, following the standardized pipeline.²⁵ Data were first normalized and scaled after basic filtering for minimum gene and cell observation frequency cut-offs. We then examined the data and performed further filtering (nFeature 7500; nCount 100 000; and percent_mt 0.1). The removal of further technical artifacts was performed using regression methods to reduce noise.

After the quality control procedures were complete, we further performed linear dimensional reduction, calculating principal components using the most variably expressed genes in our data set. Significant principal components for downstream analyses were performed similar to the methods previously reported by Macosko et al.²⁶ Cells were grouped into an optimal number of clusters for de novo cell type discovery using Seurat's FindNeighbors and FindClusters functions. Graph-based clustering approaches with visualization of cells were achieved through the use of manifold learning technique Uniform Manifold Approximation and Projection, which reduces the information captured into 2 dimensions. In total, 8861 and 3393 cells from normal skin and aGVHD skin, respectively, were used for downstream analyses.

Our analysis revealed a total of 24 clusters in the epidermis, which includes 11 keratinocyte clusters visualized using Uniform Manifold Approximation and Projection. Keratinocyte clusters (0, 1, 2, 3, 8, 9, 10, 11, 12, 15, and 18) were further divided into distinct subtypes, including basal, suprabasal, and terminal keratinocytes. Terminal keratinocytes were characterized by the expression of involucrin (*IVL*), loricrin (*LOR*), and filaggrin (*FLG*). Differentially expressed genes for a particular cell type were defined by a P value threshold $<.05$ and log fold change >0.25 . Data were subjected to Ingenuity Pathway Analysis (Qiagen, Redwood City, CA) to identify the top differentially expressed genes.

Results

Skin microbiome characteristics among allogeneic HSCT patients

The skin microbiota among our HSCT patients was enriched for predicted pathobionts, gram-negative bacteria, fungi, and viruses, from before transplant and throughout the 90-day post-HSCT period compared with the microbiome expected in healthy individuals⁹ (Figure 1A). The skin microbiome composition among our HSCT cohort was different from a previously published data set from healthy volunteers, making direct comparisons difficult (supplemental Figure 1).⁹ Bray-Curtis dissimilarity index (BCI) demonstrates separate clustering of microbial communities from HSCT patients and control healthy caregivers (supplemental Figure 2A). There was no separation of clusters when HSCT patients were stratified based on diagnosis or age (supplemental Figure 2B-C). Taxa of potential pathobionts such as beta papillomaviruses, actinomyces, and gram-negative bacteria are enriched in the skin of our HSCT patients before HSCT (supplemental Figure 2D). There was no association between skin microbiome alpha diversity (within-sample diversity; Shannon diversity index) and 1-year mortality or the development of aGVHD at any time point (Figure 1B). We calculated the BCI (0 = identical species composition and abundance; 1 = completely dissimilar) between the pretransplant microbiome and other time points to describe the stability of each individual HSCT patients' microbiome across the posttransplant recovery period. The average BCI comparing the skin microbiota of patients using the forearm specimens across the 90-day post-HSCT recovery period with their pretransplant microbiota was significantly lower among patients who survived the first year after HSCT than those who died ($P = .046$; Figure 1C). This suggests that patients who survived 1 year after HSCT had a forearm skin microbiome composition and diversity that was more similar to their pretransplant skin microbiome than those who died, but it is not clear what the clinical significance of this is. There was no significant difference in the BCI between those who developed aGVHD and those who did not develop aGVHD.

Comparing between individuals, the BCI of the arm microbiome across the 90-day posttransplant period was lower among patients who survived than those who did not ($P < 1.0 \times 10^{-20}$; Figure 1D). The pre-HSCT arm microbiome BCI among patients who survived was half the BCI among patients who died (0.47 vs 0.93; $P = 1.1 \times 10^{-5}$; Figure 1D) and was also significantly lower than healthy controls and patients who survived (0.95; $P = 4.0 \times 10^{-7}$), healthy controls alone (0.92; $P = 5.0 \times 10^{-4}$), and healthy controls and patients who died (0.98; $P = 9.0 \times 10^{-6}$; Figure 1E). We observed differential expressions of bacteria and viruses but not fungi (supplemental Figure 3), based on our outcome.

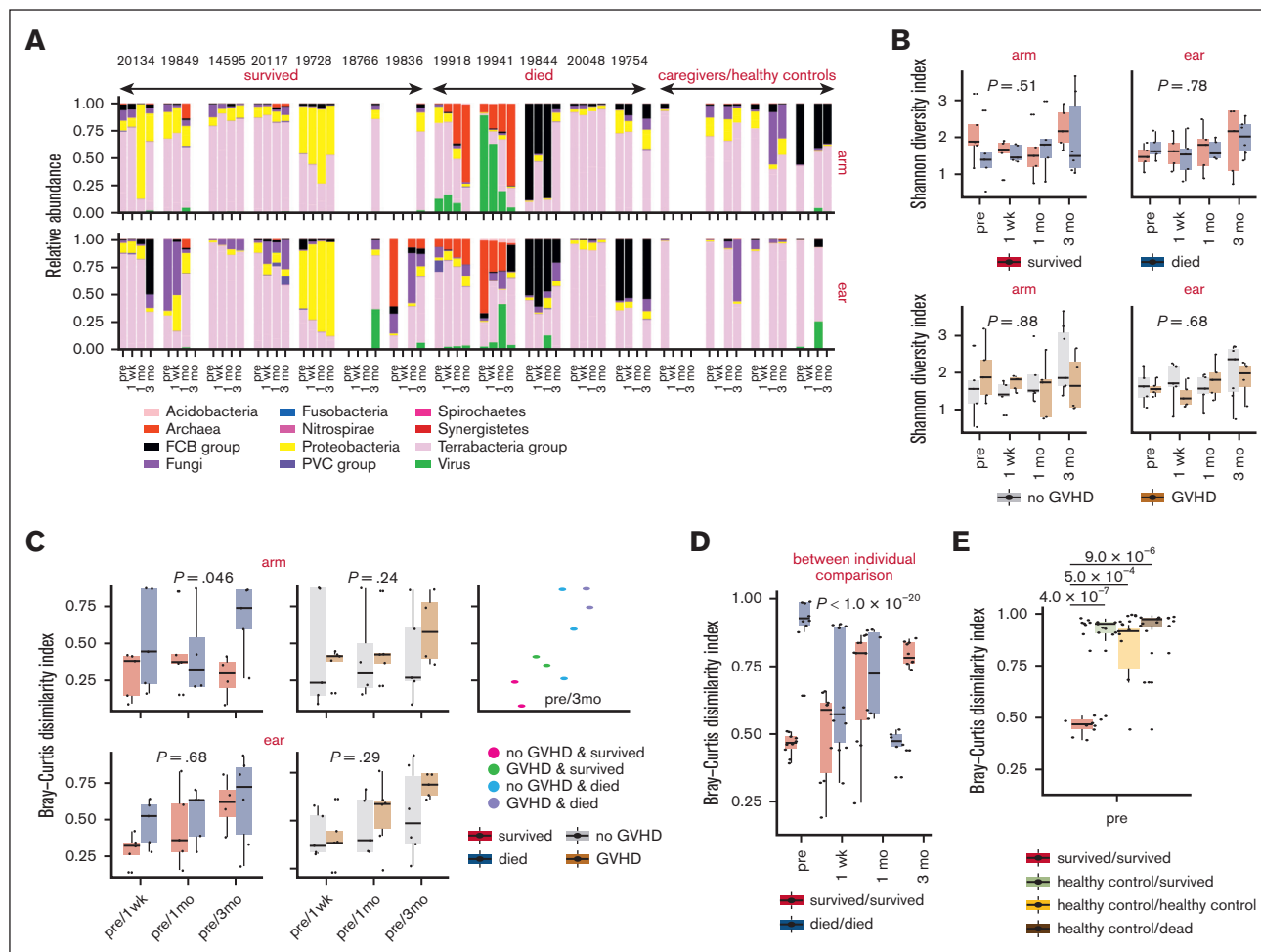


Figure 1. Skin microbiome composition across time and microbial characteristics associated with 1-year mortality and aGVHD. (A) Microbial relative abundance over time across skin sites among HSCT patients, grouped according to 1-year mortality. (B) Community diversity using Shannon diversity index and correlation with survival and aGVHD. Statistical significance based on adjusted P values for unrestricted permutation are displayed. (C) Estimation of community stability within individuals using the BCI and association with 1-year mortality and aGVHD. An index of 0 means both samples share the same species at exact the same abundances; 1 means both samples have completely different species abundance. (D) Bray-Curtis dissimilarity in forearm microbial community between individuals. Statistical significance based on adjusted P values for unrestricted permutation are displayed. (E) Between-group forearm microbial community stability at pre-HSCT time point. Healthy controls represent caregiver controls. Two-sided Wilcoxon rank sum tests were used to test for significant differences between groups, and false discovery rate-adjusted P values are shown. FCB, fibrobacterota-chlorobiota-bacteroidota; mo, months; PVC, planctomycetes-verrucomicrobia-chlamydiae; wk, weeks.

Bacteria

Analysis of the arm microbiome across our entire observation period demonstrated that the delta/epsilon and gamma classes of Proteobacteria, a major phylum of gram-negative bacteria, which includes *Escherichia coli* and *Pseudomonas*, were enriched in patients who survived compared with those who died ($P = .022$; Figure 2A). No differential expression was observed among those who developed aGVHD of the skin. We investigated a possible role of ARGs as an explanation for microbiome stability among 1-year survivors. We identified and classified patients according to the presence of plasmid-borne ARGs present at the pre-HSCT time point (Figure 2B). Patients clustered based on survival, primarily driven by fosmidomycin resistance genes and unclassified classes of ARGs.

Because commensals are important in the immunologic homeostasis of the skin, we estimated the growth rate of individual *C*

acnes strains using strain-level metagenomic estimation of growth rate.¹⁹ Starting at 1 month after HSCT, the growth rate among patients who developed aGVHD was significantly lower than those who did not ($P = .016$; Figure 2C). By 90 days after HSCT, patients in the aGVHD group had lost *C. acnes* strains, specifically strains belonging to cluster 1 (Figure 2D; supplemental Figure 4; supplemental Table 3).

Viruses

We investigated the relative abundance of viruses in the skin microbiome of our HSCT patients. Patients who did not survive the 1-year post-HSCT period had an enrichment of bacteriophages on the arm, specifically *Proteobacteria* phages but not eukaryotic viruses ($P = .012$; Figure 3A). We did not find a significant relationship between the presence of phages or eukaryotic viruses and aGVHD (Figure 3B). There was a positive correlation between

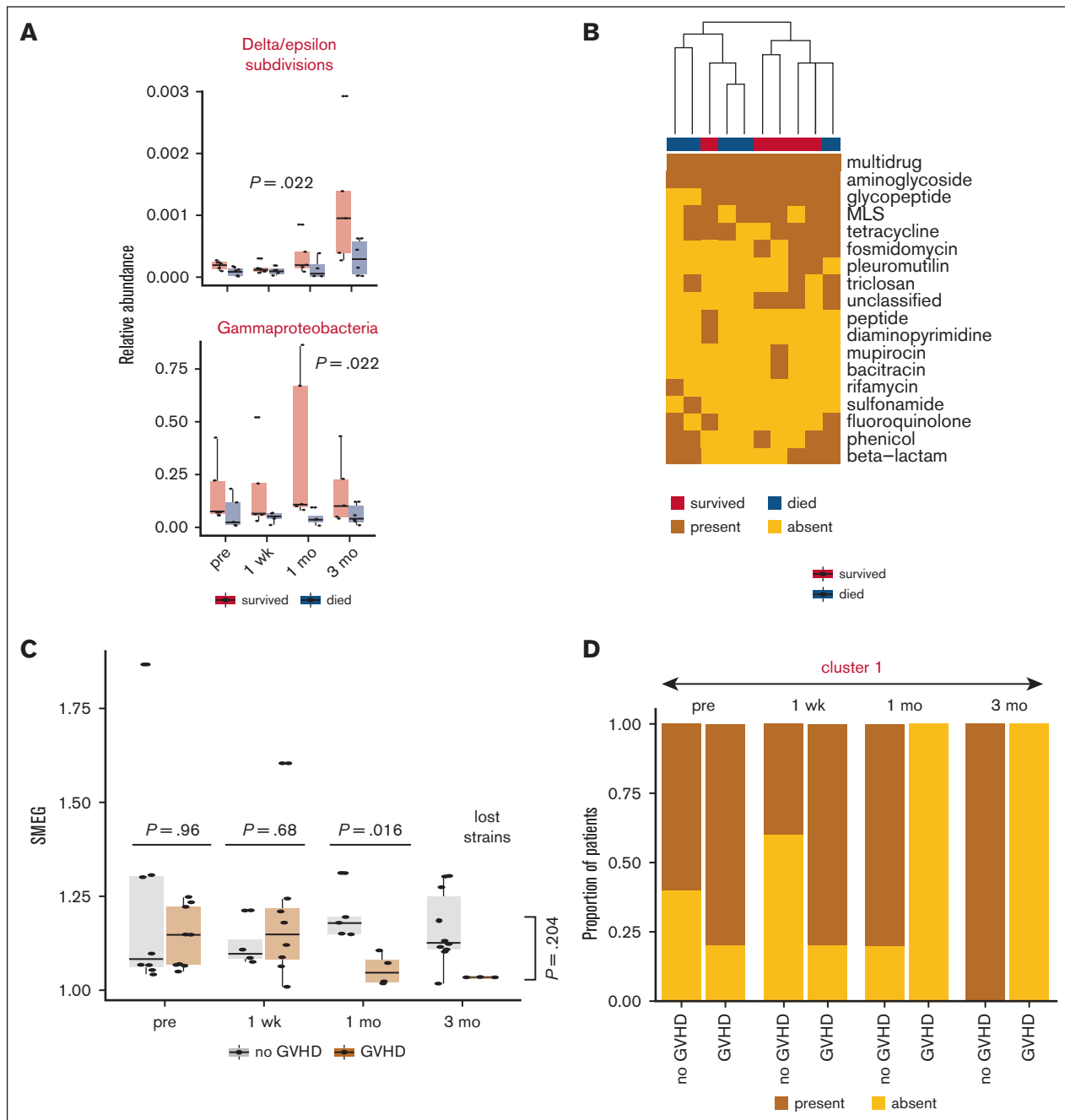


Figure 2. Bacterial community characteristics and association with 1-year mortality and acute GVHD. (A) Bacterial classes significantly associated with survival 1 year after HSCT from forearm skin sites. Statistical significance based on adjusted P values for unrestricted permutation are displayed. (B) Hierarchical clustering of plasmid-borne ARGs from forearm skin site at pre-HSCT time point. (C) Estimation of subspecies/strain-level growth rate for *C. acnes* strains and association with aGVHD from forearm skin site. A higher SMEG score indicates a higher growth rate. Each data point represents a subspecies cluster. Significant differences at pretransplant, 1-week, 1-month, and 3-month time points were computed with Wilcoxon rank sum test. Overall P value ($P = .204$) between groups, accounting for multiple measures was computed based on unrestricted permutation. (D) Distribution of a *C. acnes* cluster over time as a function of aGVHD from forearm skin sites. MLS, macrolides, lincosamides, streptogramins; mo, months; SMEG, strain-level metagenomic estimation of growth rate; wk, weeks.

phages and delta/epsilon *Proteobacteria* at week 1 but a negative correlation at 3 months in the arm skin microbiome among patients who died before 1 year after HSCT (Figure 3C).

Because polyomaviruses have been linked to cutaneous tumors and eruptions among immunosuppressed patients,²⁷ we

investigated the relative abundance of polyomaviruses in the skin microbiome among our cohort. The relative abundance of polyomaviruses in both the arm and retroauricular skin microbiome was significantly higher among patients who died than those who survived the 1-year post-HSCT period (arm, $P = .006$; ear, $P = 2 \times 10^{-16}$; Figure 4A). Several of polyomavirus strains were identified

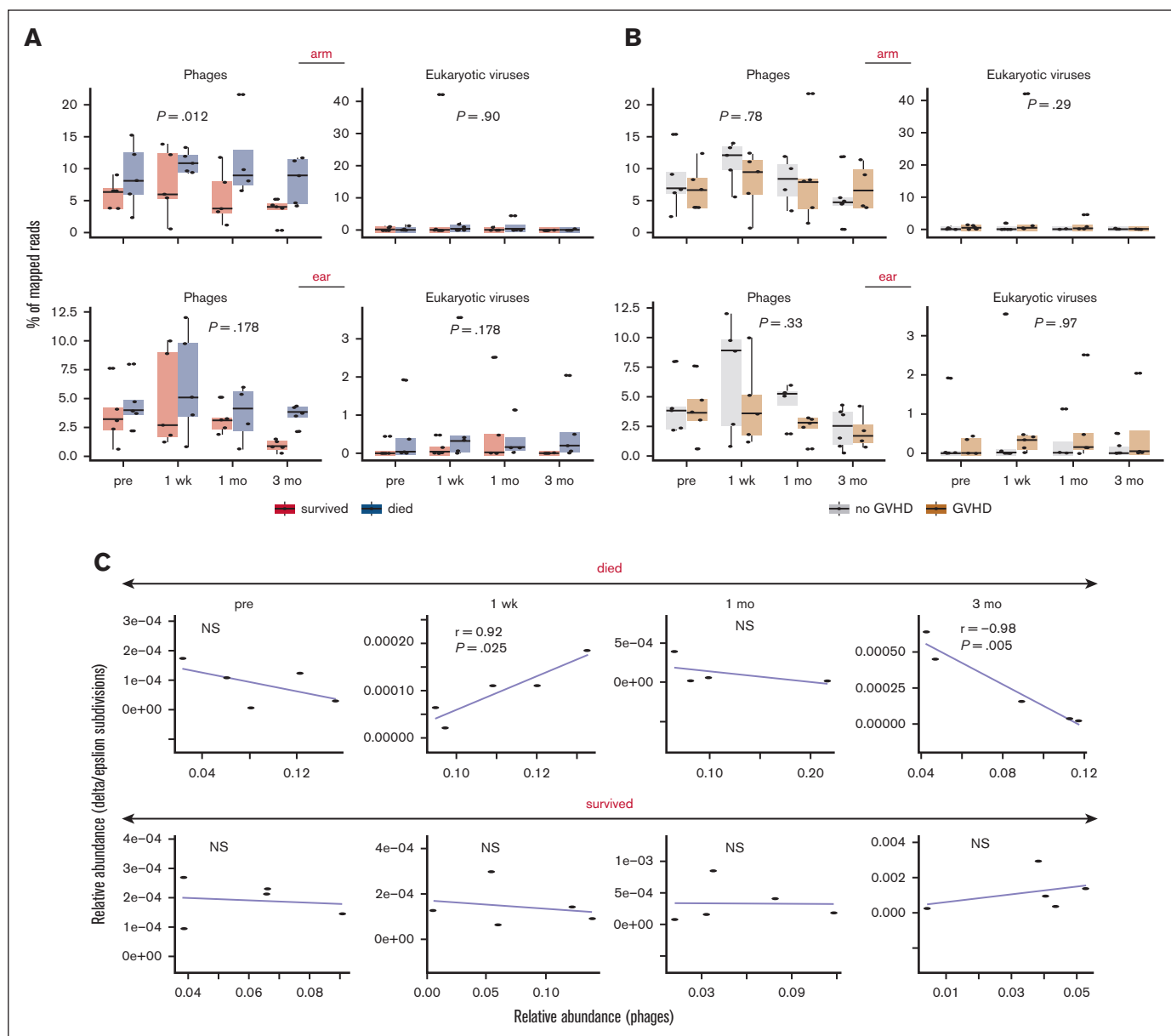


Figure 3. Phage and eukaryotic virus association with survival and aGVHD. (A-B) proportion of phage and eukaryotic virus reads and association with survival 1 year after HSCT (A) and aGVHD (B). Statistical significance based on adjusted P values for unrestricted permutation are displayed. (C) Pearson correlation between relative abundances of delta/epsilon divisions *Proteobacteria* and bacteriophages from the forearm skin site. mo, months; NS, nonsignificant (ie, $P > .05$); wk, weeks.

among patients who died before 1 year after HSCT (Figure 4B). As proof of concept to identify whether polyomavirus strains were shared between patient and caregiver, we used single-nucleotide polymorphism analysis to compare polyomavirus genomes. We identified shared polyomavirus strains between 1 caregiver and patient pair (Figure 4C; supplemental Figure 5). Overall, polyomavirus strain composition tended to be individual specific and temporally stable over different time points (Figure 4D).

Keratinocyte response in aGVHD

Using scRNA-seq, we identified 24 clusters of cells from the epidermis obtained from lesional skin with new onset aGVHD and a healthy volunteer (Figure 5A). We distinguished terminally

differentiated keratinocytes among total keratinocytes based on their expression of involucrin (*IVL*), loricrin (*LOR*), and filaggrin (*FLG*). The top 3 differentially expressed genes in total keratinocytes (Figure 5B) and terminal keratinocytes (Figure 5C) in aGVHD skin compared with cells obtained from the skin of a healthy volunteer were the antimicrobial peptides S100A8, S100A9, and S100A7 (supplemental Table 5). Elafin (peptidase inhibitor 3/PI3), a serine protease inhibitor with antimicrobial properties, which has been proposed as both a biomarker for cutaneous aGVHD²⁸ and a predictor indicator of poorer outcomes,²⁹ was the fourth most highly differentially expressed genes in both keratinocytes and terminal keratinocytes. Interleukin-36 γ , whose expression is strongly induced by microbial triggers and whose overexpression is

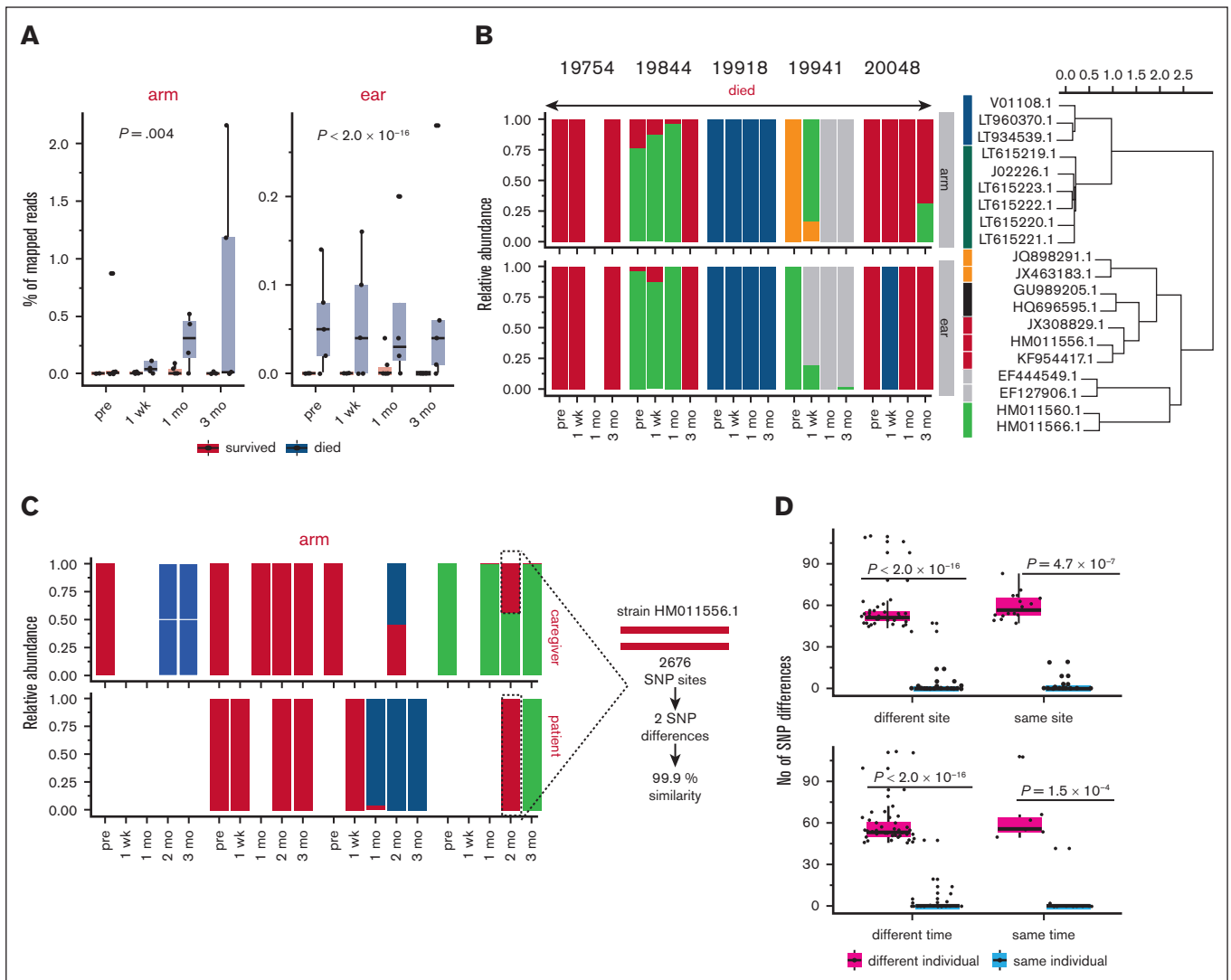


Figure 4. Polyomavirus strain distribution and transmission between skin sites and individuals. (A) Proportion of polyomavirus associated with survival 1 year after HSCT. Statistical significance based on adjusted P values for unrestricted permutation are displayed. (B) Relative abundance of polyomavirus strains over time and across skin sites. (C) Polyomavirus strain distribution between matched healthy caregiver–patient pair in the forearm skin site. Transmission of strain HM011556.1 was analyzed by identifying differences in single-nucleotide polymorphisms (SNPs) between a matched caregiver and patient pair. (D) Between- and within-individual polyomavirus strain (HM011556.1) similarity. Each data point represents a sample pair. Higher SNP differences represent more dissimilarity. Two-sided Wilcoxon rank sum tests were used to test for significant differences between groups. mo, months; wk, weeks.

linked to the immunopathology of psoriasis,³⁰ was strongly differentially expressed, as were keratins 6a and 16, which are induced in response to epidermal barrier stress³¹ (Figure 5B-C).

Discussion

The skin microbiome has a profound effect on skin immunity, and many protective aspects of this immune modulation are compartmentalized from systemic immune effects.^{32,33} Disruption of skin microbiota is intimately associated with inflammatory skin diseases such as eczema and psoriasis, and this process is exacerbated by epidermal barrier failure.³⁴ Several factors could contribute to skin dysbiosis among allogeneic HSCT patients. Disruption in the local skin environment, expected after radiation and chemotherapy, such

as changes in sebocyte activity and moisture of the skin, as well as systemic immune deficiency, can alter the landscape of the cutaneous microbiome composition.^{35,36} The ubiquitous presence of pathogenic microbes, which are known triggers of the innate immune response, coupled with an increased keratinocyte antimicrobial inflammatory response in patients with aGVHD, suggest a role for the host–microbiome interaction in either the initiation or propagation of cutaneous aGVHD. Skin commensals are important for the development of the local, regulatory immune compartment of the skin.⁸ On the contrary, some authors have observed that commensals can increase skin inflammation and impair wound healing.³⁷ *C. acnes* is part of the normal flora of the skin. Because of the limitations of this study, we cannot draw any conclusions regarding either the cause or impact of the loss of *C. acnes* in the

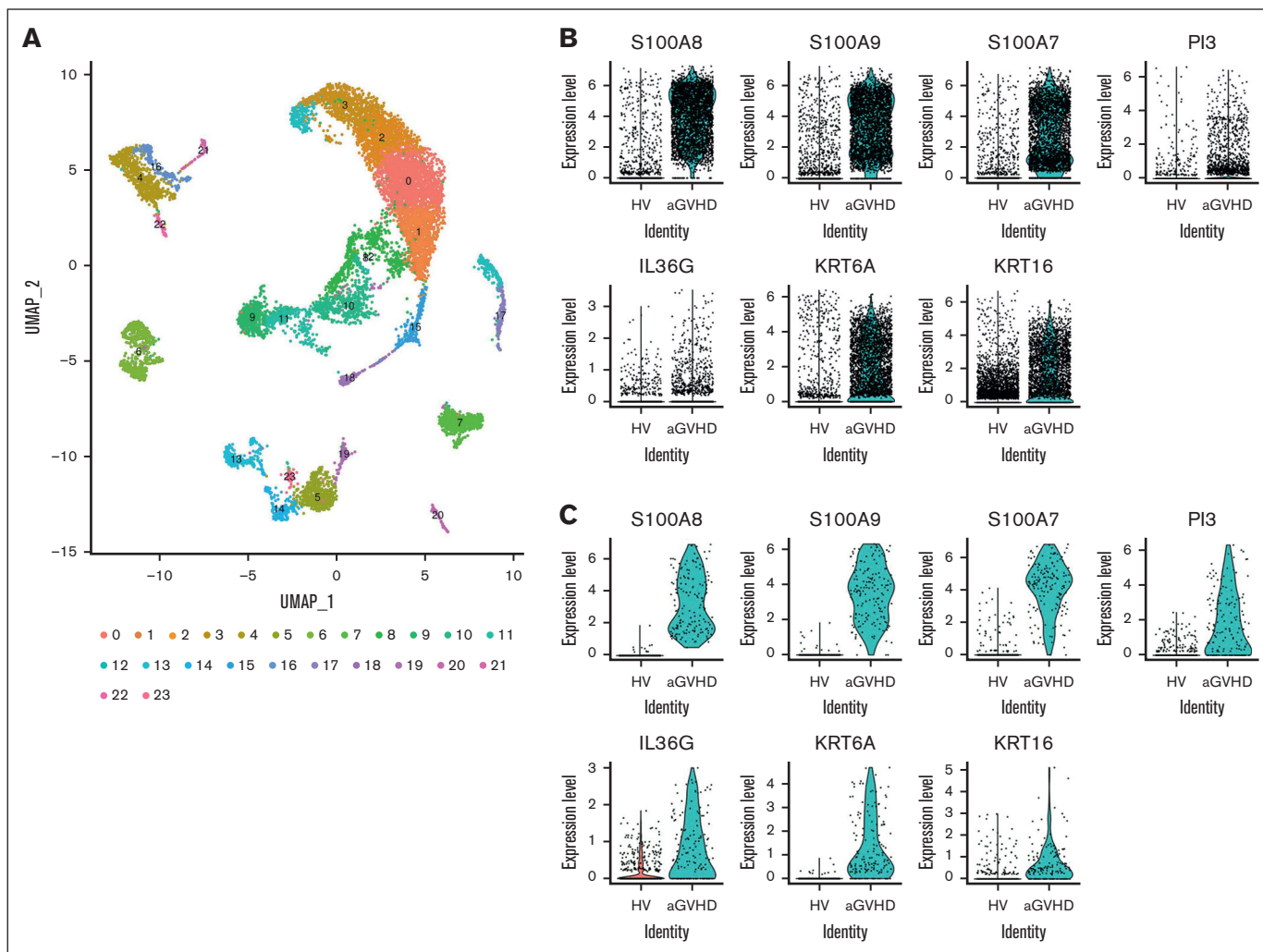


Figure 5. Single-cell RNA-sequencing of epidermal keratinocytes demonstrates the expression of inflammatory and proliferative markers in acute cutaneous GVHD. (A) UMAP plot of scRNA-seq shows unsupervised clusters of 8861 and 3393 cells isolated from normal skin from a healthy volunteer (HV) and aGVHD epidermis samples, respectively, including 11 keratinocyte clusters (0, 1, 2, 3, 8, 9, 10, 11, 12, 15, and 18). (B) Violin plots show expression levels of S100A8, S100A9, S100A7, *PI3*, *IL36G*, *KRT6A*, and *KRT16* in all epidermal cells in normal HV skin vs aGVHD samples. (C) Violin plots show expression levels of S100A8, S100A9, S100A7, *PI3*, *IL36G*, *KRT6A*, and *KRT16* in terminally differentiated keratinocytes in normal HV skin vs aGVHD samples. UMAP, Uniform Manifold Approximation and Projection.

post-HSCT period. Larger, prospective studies that can correlate host changes such as the disruption skin barrier integrity and environment, changes in local and systemic immune compartments, and antibiotic use are needed. It remains to be seen what the overall impact of the cutaneous microbiome is on the local immune homeostasis and reconstitution after allogeneic HSCT.

We observed that patients who survived were more likely to maintain their pre-HSCT microbiome composition and diversity in the 3-month posttransplant recovery period, although there may be some fluctuation. In our cohort, the skin microbiome composition among patients who survived were more similar to each other before transplant than those who did not survive or healthy/caregiver controls. In addition, the microbial community composition of those who survived was dissimilar from those of healthy/caregiver controls. The exact composition of this "survival signature" of the skin microbiome, which seems to be evident even in the pre-transplant period, and the immunologic function or status that it

reflects need to be explored further. Because systemic immunodeficiency can alter skin microbiome composition, the changes we observed may be a reflection of the patient's overall immune status. Infection is a leading cause of non-relapse-related mortality among allogeneic HSCT patients, and the skin may also serve as a reservoir of pathogens.

We noted an inverse relationship between the enrichment of delta/epsilon *Proteobacteria* in survivors on one hand and the increased presence of bacteriophages in the skin microbiome of those who did not survive on the other. We inferred that the bacteriophages may be in the early, nonvirulent lysogenic phase at week 1 and in the virulent lytic phase at 3 months, and the enrichment of delta/epsilon *Proteobacteria* in survivors may be attributed, partly, to the absence of lytic phages.³⁸

Our observation that skin microbiome instability and increased viral load may be associated with 1-year mortality after HSCT implies a

relationship between systemic immune status and the skin microbiome homeostasis. In healthy individuals, the skin microbiome at most skin sites is relatively stable over time.⁹ This stability may be an inherent characteristic of a healthier community. This is despite the fact that the configuration of the pre-HSCT skin microbiome is not normally observed among healthy individuals but is common among the pre-HSCT patients. All this suggests that stability, rather than a recapitulation of a “healthy” microbiome, may be an important biomarker for survival. A shifting skin microbiome may be a sign of worsening skin barrier or deteriorating immune surveillance, but this remains to be proven. The impact of skin barrier integrity in the development of aGVHD deserves additional studies. The impact of antibiotic and antiseptic use and the post-HSCT recovery environment on dysbiosis also need further exploration.

Our study is limited by the small number of patients and controls and the availability of sampling from other anatomic sites. In addition, it is difficult to standardize antibiotic exposure and aGVHD treatments among HSCT patients. Nevertheless, larger prospective studies are warranted. The relationship between the skin microbiome and gut microbiome may also be worth examining. Non-antibiotic-based interventions, such as changing the patient's post-HSCT care to a home rather than a hospital-based environment, could potentially affect the skin microbiome and patient outcomes.

Acknowledgments

The authors acknowledge the assistance of the Duke Molecular Physiology Institute Molecular Genomics Core for the generation of single-cell RNA sequencing data for the manuscript.

This study was supported by the Duke Medicine Scholars award (A.R.C.), National Cancer Institute (NCI) grant R21 CA223419 (A.R.C. and J.Z.), NCI grant R01CA203950 (A.D.S., L.M.B., and N.J.C.), National Institute on Aging (NIA) grant R21AG066388 (A.D.S., L.M.B., and N.J.C.), the American Society of Hematology Scholar award (A.D.S.), and Duke Pepper Center/

NIA grant P30-AG028716-13 Mini#6 (A.D.S.). J.O. is supported by the National Institutes of Health (NIH) grants DP2 GM126893-01, 1U54NS105539, 1 U19 AI142733, and 1 R21 AR075174, National Science Foundation grant 1853071, the American Cancer Society, and the Leo Foundation. A.E. is supported by NIH grant K99GM135540, the Jackson Laboratory Scholars Program, and the Maximilian & Marion Hoffman Foundation Postdoctoral fellow.

The funders had no role in the conceptualization, design, data collection, analysis, decision to publish, or preparation of the manuscript.

Authorship

Contribution: A.R.C. and J.O. conceptualized, designed, and supervised the article; N.J.C., R.P.H., and A.D.S. designed and supervised critical aspects of this study; A.R.C., A.D.S., N.J.C., and J.O. obtained funding; A.R.C., C.P., K.R.N., A.D.S., E.F., J.Z., Y.J.J., M.V.L., L.M.B., and A.T.B. collected the data; A.R.C., A.E., R.P.H., E.F., J.O., A.D.S., A.J.P., V.J., S.G.G., and J.Z. analyzed and interpreted the data; A.R.C., A.E., and A.J.P. wrote the original draft of the manuscript; and A.R.C., A.E., R.P.H., A.D.S., J.Z., N.J.C., K.M.S., S.G.G., and J.O. reviewed the manuscript and provided substantial contribution to the analysis and interpretation of the data and editing of the manuscript.

Conflict-of-interest disclosure: The authors declare no competing financial interests.

ORCID profiles: A.R.C., 0000-0001-7489-2046; J.Z., 0000-0002-4485-1750; A.T.B., 0000-0002-4495-3765; E.F., 0000-0003-0588-5066; K.M.S., 0000-0002-1379-9216; N.J.C., 0000-0001-6725-7220; J.O., 0000-0002-5534-7023.

Correspondence: Adela R. Cardones, Division of Dermatology, Department of Internal Medicine, University of Kansas Medical Center, 3901 Rainbow Blvd, Delp 4094, Mailstop 2025, Kansas City, KS 66160; email: acardones@kumc.edu.

References

- Martin PJ, Schoch G, Fisher L, et al. A retrospective analysis of therapy for acute graft-versus-host disease: initial treatment. *Blood*. 1990;76(8):1464-1472.
- Schwabkey ZI, Jenq RR. Microbiome anomalies in allogeneic hematopoietic cell transplantation. *Annu Rev Med*. 2020;71:137-148.
- Peled JU, Gomes ALC, Devlin SM, et al. Microbiota as predictor of mortality in allogeneic hematopoietic-cell transplantation. *N Engl J Med*. 2020;382(9):822-834.
- Oh J, Freeman AF, et al; NISC Comparative Sequencing Program. The altered landscape of the human skin microbiome in patients with primary immunodeficiencies. *Genome Res*. 2013;23(12):2103-2114.
- Kong HH, Oh J, Deming C, et al. Temporal shifts in the skin microbiome associated with disease flares and treatment in children with atopic dermatitis. *Genome Res*. 2012;22(5):850-859.
- Chang HW, Yan D, Singh R, et al. Alteration of the cutaneous microbiome in psoriasis and potential role in Th17 polarization. *Microbiome*. 2018;6(1):154.
- Nakatsuji T, Chen TH, Butcher AM, et al. A commensal strain of *Staphylococcus epidermidis* protects against skin neoplasia. *Sci Adv*. 2018;4(2):eaao4502.
- Scharschmidt TC, Vasquez KS, Truong HA, et al. A wave of regulatory T cells into neonatal skin mediates tolerance to commensal microbes. *Immunity*. 2015;43(5):1011-1021.
- Oh J, Byrd AL, Park M, Kong HH, Segre JA; NISC Comparative Sequencing Program. Temporal stability of the human skin microbiome. *Cell*. 2016;165(4):854-866.

10. Trim Galore. Babraham Bioinformatics. 2015. Accessed 1 January 2021. <https://github.com/FelixKrueger/TrimGalore>
11. Langmead B, Salzberg SL. Fast gapped-read alignment with Bowtie 2. *Nat Methods*. 2012;9(4):357-359.
12. Joshi NA, Fass JN. Sickle: A sliding-window, adaptive, quality-based trimming tool for FastQ files (Version 1.33). 2011. Accessed 1 January 2021. <https://github.com/najoshi/sickle>
13. Zhou W, Gay N, Oh J. ReprDB and panDB: minimalist databases with maximal microbial representation. *Microbiome*. 2018;6(1):15.
14. Hong C, Manimaran S, Shen Y, et al. PathoScope 2.0: a complete computational framework for strain identification in environmental or clinical sequencing samples. *Microbiome*. 2014;2:33.
15. Li D, Liu CM, Luo R, Sadakane K, Lam TW. MEGAHIT: an ultra-fast single-node solution for large and complex metagenomics assembly via succinct de Bruijn graph. *Bioinformatics*. 2015;31(10):1674-1676.
16. Krawczyk PS, Lipinski L, Dziembowski A. PlasFlow: predicting plasmid sequences in metagenomic data using genome signatures. *Nucleic Acids Res*. 2018;46(6):e35.
17. Hyatt D, Chen GL, Locascio PF, Land ML, Larimer FW, Hauser LJ. Prodigal: prokaryotic gene recognition and translation initiation site identification. *BMC Bioinf*. 2010;11:119.
18. Arango-Argoty G, Garner E, Pruden A, Heath LS, Vikesland P, Zhang L. DeepARG: a deep learning approach for predicting antibiotic resistance genes from metagenomic data. *Microbiome*. 2018;6(1):23.
19. Emiola A, Zhou W, Oh J. Metagenomic growth rate inferences of strains in situ. *Sci Adv*. 2020;6(17):eaaz2299.
20. Paez-Espino D, Chen IMA, Palaniappan K, et al. IMG/VR: a database of cultured and uncultured DNA viruses and retroviruses. *Nucleic Acids Res*. 2017; 45(D1):D457-D465.
21. Hampras SS, Giuliano AR, Lin HY, et al. Natural history of polyomaviruses in men: the HPV infection in men (HIM) study. *J Infect Dis*. 2015;211(9): 1437-1446.
22. Price MN, Dehal PS, Arkin AP. FastTree 2—approximately maximum-likelihood trees for large alignments. *PLoS One*. 2010;5(3):e9490.
23. Katoh K, Standley DM. MAFFT multiple sequence alignment software version 7: improvements in performance and usability. *Mol Biol Evol*. 2013;30(4): 772-780.
24. Li H, Handsaker B, Wysoker A, et al. The sequence alignment/map format and SAMtools. *Bioinformatics*. 2009;25(16):2078-2079.
25. Butler A, Hoffman P, Smibert P, Papalexi E, Satija R. Integrating single-cell transcriptomic data across different conditions, technologies, and species. *Nat Biotechnol*. 2018;36(5):411-420.
26. Macosko EZ, Basu A, Satija R, et al. Highly parallel genome-wide expression profiling of individual cells using nanoliter droplets. *Cell*. 2015;161(5): 1202-1214.
27. Sheu JC, Tran J, Rady PL, Dao H Jr, Tyring SK, Nguyen HP. Polyomaviruses of the skin: integrating molecular and clinical advances in an emerging class of viruses. *Br J Dermatol*. 2019;180(6):1302-1311.
28. Paczesny S, Braun TM, Levine JE, et al. Elafin is a biomarker of graft-versus-host disease of the skin. *Sci Transl Med*. 2010;2(13):13ra2.
29. Bruggen MC, Petzelbauer P, Greinix H, et al. Epidermal elafin expression is an indicator of poor prognosis in cutaneous graft-versus-host disease. *J Invest Dermatol*. 2015;135(4):999-1006.
30. Bridgwood C, Fearnley GW, Berekermeri A, et al. IL-36 γ is a strong inducer of IL-23 in psoriatic cells and activates angiogenesis. *Front Immunol*. 2018; 9:200.
31. Lessard JC, Pina-Paz S, Rotty JD, et al. Keratin 16 regulates innate immunity in response to epidermal barrier breach. *Proc Natl Acad Sci U S A*. 2013; 110(48):19537-19542.
32. Chen YE, Fischbach MA, Belkaid Y. Skin microbiota-host interactions. *Nature*. 2018;553(7689):427-436.
33. Linehan JL, Harrison OJ, Han SJ, et al. Non-classical immunity controls microbiota impact on skin immunity and tissue repair. *Cell*. 2018;172(4):784-796.e18.
34. Kobayashi T, Glatz M, Horiuchi K, et al. Dysbiosis and staphylococcus aureus colonization drives inflammation in atopic dermatitis. *Immunity*. 2015; 42(4):756-766.
35. Abusleme L, Diaz PI, Freeman AF, et al. Human defects in STAT3 promote oral mucosal fungal and bacterial dysbiosis. *JCI Insight*. 2018;3(17): e122061.
36. Howard B, Bascom CC, Hu P, et al. Aging-associated changes in the adult human skin microbiome and the host factors that affect skin microbiome composition. *J Invest Dermatol*. 2022;142(7):1934-1946.e21.
37. Khadka VD, Markey L, Boucher M, Lieberman TD. Commensal skin bacteria exacerbate inflammation and delay skin barrier repair. *J Invest Dermatol*. 2024;144(11):2541-2552.e10.
38. Silveira CB, Rohwer FL. Piggyback-the-winner in host-associated microbial communities. *NPJ Biofilms Microbiomes*. 2016;2:16010.